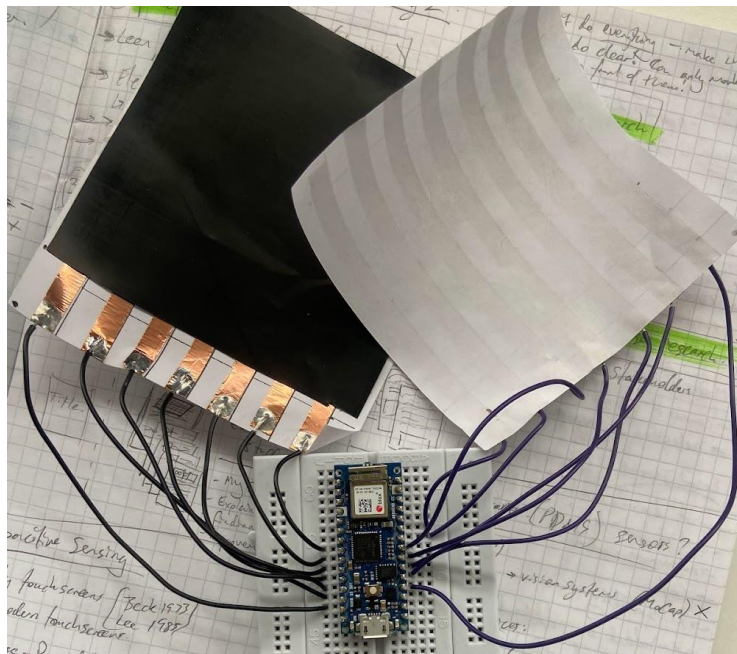


Mini-Study 2: Initial Prototyping



Design of a Pressure Sensing Array for the Inflation Prone Repositioning Device

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Word count: 2746

04-Dec-2023

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Abbreviations

IPRD	Inflation Proning Repositioning Device
ARDS	Acute Respiratory Distress Syndrome
PP	Prone Position
PU	Pressure Ulcer
ICU	Intensive Care Unit
IDE	Integrated Development Environment
CNT	Carbon Nano-Tube
PDMS	Polydimethylsiloxane

1 Introduction

Following work presented in the first study, this report will build on developing an understanding of pressure sensor arrays in the context of the Inflation Prone Repositioning Device (IPRD) to continue development of a design specification.

This report covers initial prototyping activities to demonstrate proof-of-concept of a small-scale pressure sensor matrix. This includes its design, construction, performance and analysis to highlight areas for potential improvement and further development.

This work aligns with the design thinking methodology for creative problem solving [1] (Figure 1). This study represents one iteration of the loop by building on problem definition and ideation in the first study, to prototyping and testing to further define the problem in this report.

Specifically this report covers how the basic prototype functions and analyses the performance to identify areas that could be improved in future iterations. These focuses are discussed in relation to the user needs in the context of the IPRD project.

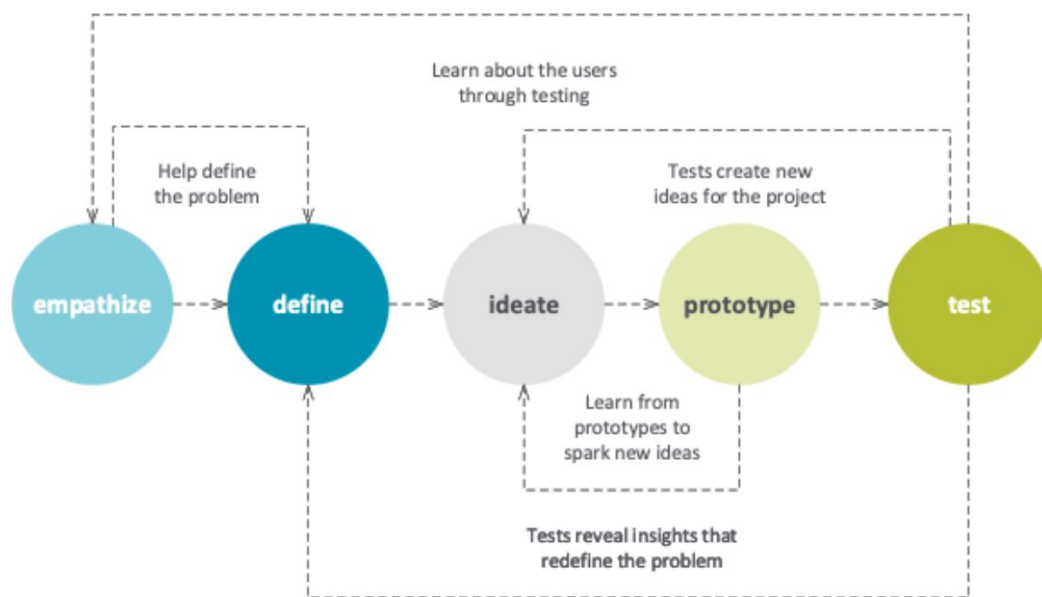


Figure 1 Design Thinking methodology[1] applied in this report. This study represents one iteration of the prototype/test/define loop

2 Initial Prototyping

To demonstrate proof-of-concept of a small piezoresistive-based sensor matrix controlled by an Arduino was constructed to test the construction, operating principles, and code required to run a pressure sensor matrix. In its most basic form a piezoresistive pressure sensor arrays can be constructed from a sheet of Velostat and copper tape.

Velostat is a polymer impregnated with carbon-black to make it electrically conductive. Polymers that are usually insulating can be made conductive by adding conductive filler (carbon black) above a certain 'percolation threshold' to create continuous conductive paths in the polymer matrix[2], [3]. Figure 2 shows a representation of this principle[4]. Elastic deformation of the material causes more particles to contact and therefore reduces resistivity when the material is compressed. The change in resistivity can be quantified with basic instrumentation electronics (potential divider or Wheatstone bridge) making it simple to implement. However, piezoresistive polymers are susceptible to hysteresis and are not the most suitable for fast-changing, dynamic loads [5].

Velostat can be purchased in small sections quite cheaply (£4 per 30x30cm sheet [6]), and is commonly used in hobby electronics and DIY applications for its cheap price and relatively simple implementation[7], [8].

Figure 3 shows conductivity versus concentration of carbon black in polymers. The steep middle section shows the percolation threshold where the polymer conductivity steeply rises for a small change in carbon concentration[4].

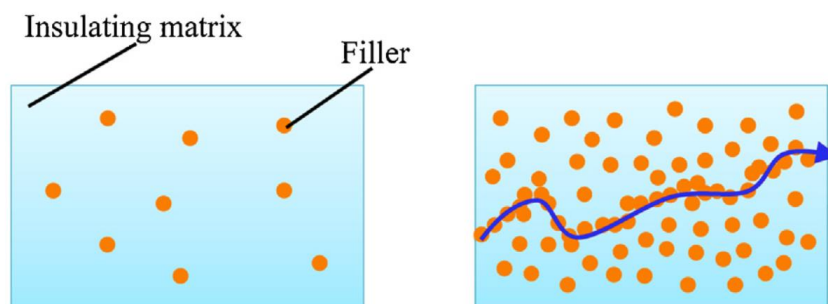


Figure 2 Representation of the mechanism of piezoresistivity in polymer[2]

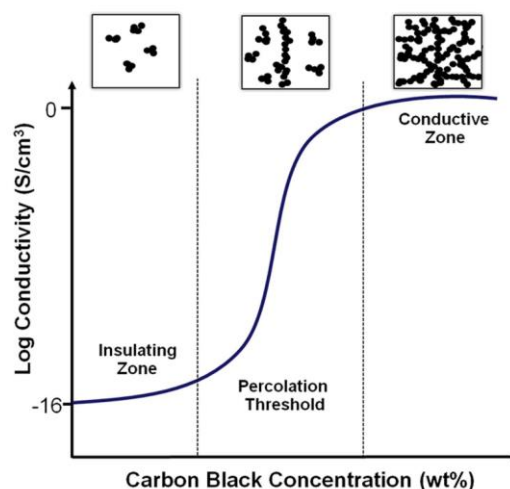


Figure 3 Graph of log conductivity against carbon black concentration to show S-shaped percolation curve[4]

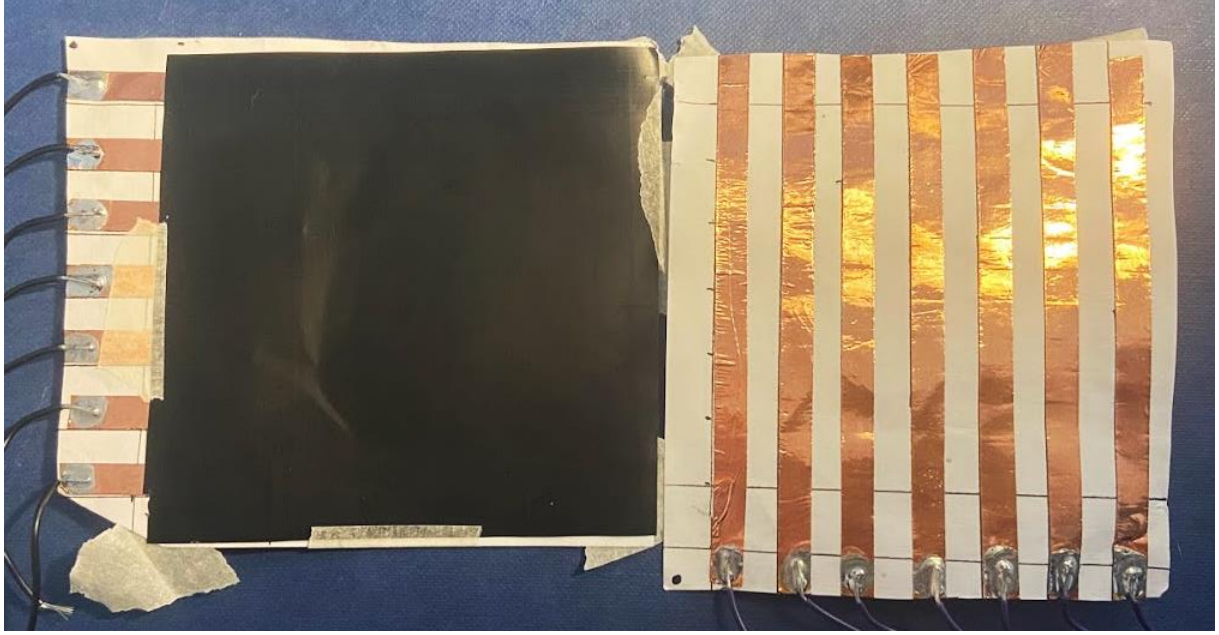


Figure 4 Construction of the initial piezoresistive pressure sensor array prototype

A 7x7 matrix was constructed using Velostat and copper tape as shown in Figure 4. The copper tape was 6mm wide, with 6mm spacing in between rows to give a sensing cell size of 12mm (resolution of 0.83 d/cm). The top layer of copper tape are the columns, and the bottom layer are the rows. A sheet of Velostat is sandwiched in between to act as a pressure varying resistance between each of the 49 intersection points of the matrix.

A simple method of driving sensor arrays is directly addressing each sensor individually; however this requires $2 \times M \times N$ connections for an array of $M \times N$ [3]. Using a passive matrix (Figure 4) requires $M+N$ connections to measure pressure at each node. As the size of the array increases this method becomes necessary to efficiently connect to and take readings from the array[9], [10].

This prototype was simply made to test the operating principles and function of a pressure sensor, so paper and masking tape were used to bond the layers together, future iterations will consider material choice more carefully.

2.1 Software

To drive the sensor array each row and column were individually connected to an Arduino Nano microcontroller (MCU). Figure 5 shows a flowchart of the software process to read the sensor array and output a live updating matrix of relative pressure.

C++ written in Arduino IDE is the main interfacing and voltage reading code.

In the Arduino script, firstly the connections to each row and column are defined. Each row is connected to a pulled-up analog pin. Figure 6 shows a simplified schematic of this to demonstrate how this creates a voltage divider using the internal Arduino pull-up resistor and the variable resistance of the Velostat[11]. The output voltage read by the Arduino is given by (1), where V_{out} is the output voltage read by the Arduino, V_{in} is 3.3V from the pulled-up pin, and R_1 and R_2 are the pullup resistor, and Velostat resistance, respectively.

$$V_{out} = v_{in} \left(\frac{R_1}{R_1 + R_2} \right) \quad (1)$$

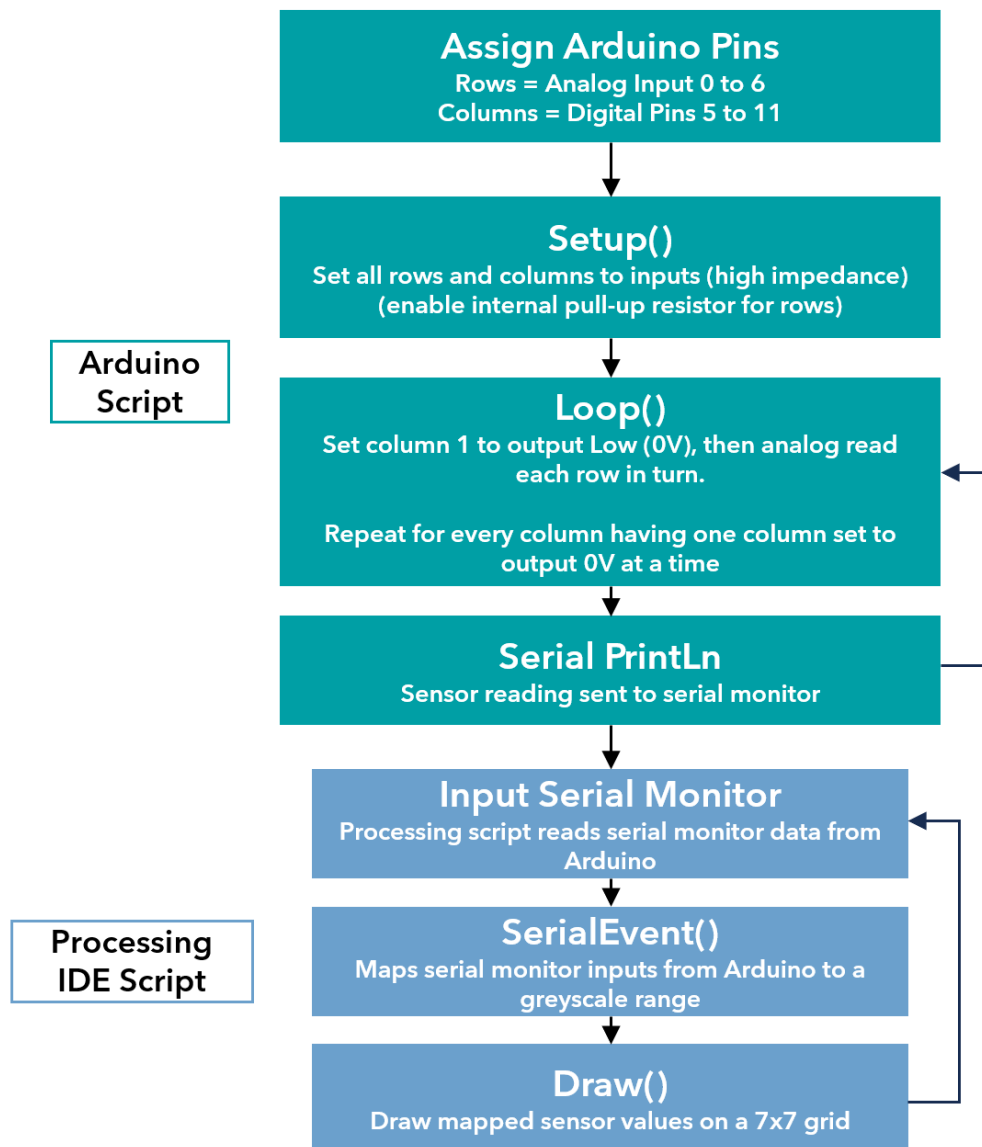


Figure 5 Flow chart of set-up and processing scripts to drive the pressure sensor array Figure 4

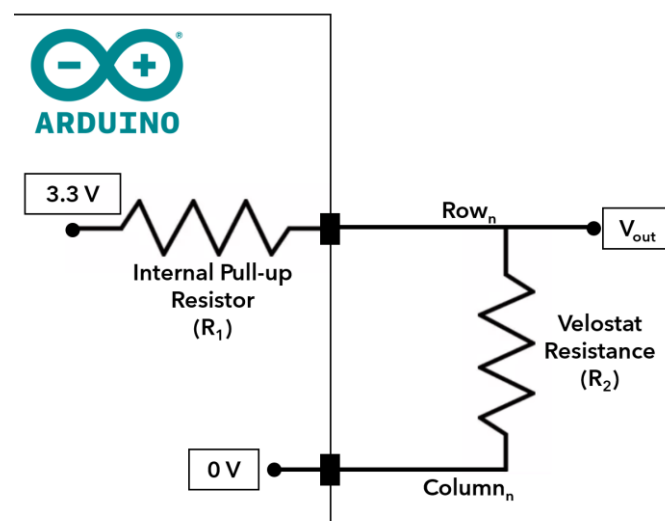


Figure 6 Simplified schematic of voltage divider setup to measure resistance changes in the Velostat

After the rows and columns are assigned to Arduino inputs and setup, the function `loop()`, iterated through each column and row to measure voltages. Figure 7 shows this loop in more detail. A column is set to LOW (0V) (the rest set to HIGH (high impedance)), and then an analog voltage reading is taken from each row. This is then repeated for every column. The voltage readings are then sent to the Serial Monitor – this is a tool which allows data from the Arduino to be sent to the visual processing program [12].

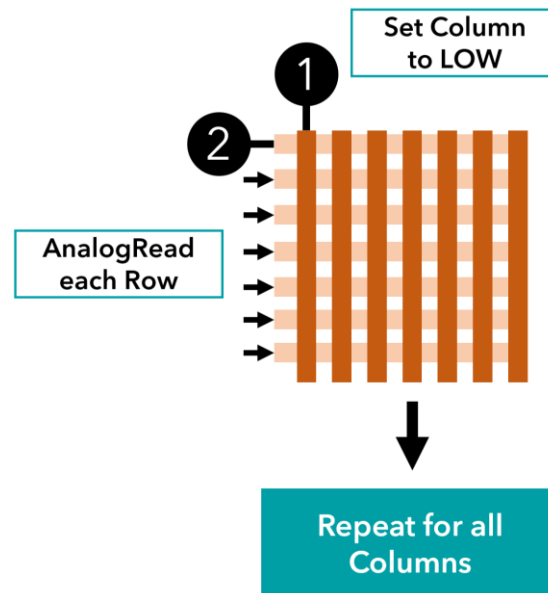


Figure 7 Visual explanation of the `loop()` function in the Arduino code

Voltage readings from the sensor are sent via the Serial Monitor to a second script written in Processing IDE, another open-source IDE that allows real-time visualisation of sensor data[13][14]. Incoming sensor readings are values from 0-1023 (due to the Arduino's on-board 10-bit ADC). The Processing script maps these to a 0-255 greyscale range, and then uses the `draw()` function to create a live updating matrix showing the pressure at each node based on incoming data.

A short video of the prototype functioning is attached as supporting information to this document (see *PrototypeV0.mp4*). Figure 8 shows a still from this video to show the pressure sensor prototype with visualisation script running.

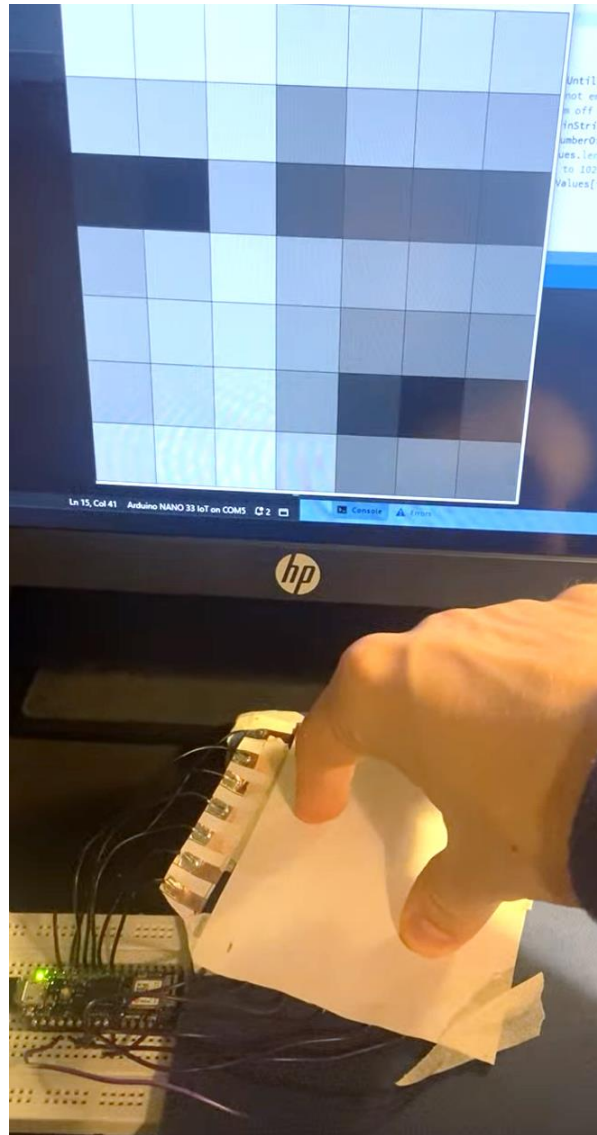


Figure 8 Image from attached video showing that the prototype sensor array can detect and locate multiple points of contact

2.2 Discussion and Improvements

The presented prototype is a very rough initial prototype, but it is a useful exercise to highlight unforeseen problems, and guide future developments.

Material selection is a clear improvement required as the final prototype will not use paper as a substrate, which was used for ease and speed of prototyping. Material selection research is covered later in Section 3, in relation to flexibility, stretch, sterilisation and other factors in the context of the IPRD project.

2.2.1 Ghosting

The attached video and Figure 9 show that the prototype suffers from 'ghosting' [2][13], where it appears the nodes adjacent in the same row also 'detect' pressure when in reality there is none present.

Ghosting is an issue also found in computer keyboards, which are analogous since they operate on a similar passive matrix array to the prototype sensor array presented. In a computer keyboard each key is connected to a row and column, pressing the key allows current to flow from the column to the row. A current is applied to each column in turn, and the controller checks to see which rows output current. A simple mapping converts combinations of row and column into a letter. [16], [17]

Ghosting occurs when current flows the wrong way through a switch (node) and causes a row to output current when it should not. Figure 10 shows an example of this where keys **w**, **e** and **d** are pressed, but **w** and **s** are outputted. When current is applied to Column 2 there is a parasitic path forwards through **w**, backwards through **e** and then through **d** that will cause both row 1 and 2 to be activated, outputting the letter **s** incorrectly.

The case of a resistive sensor is similar, just replacing switches with the variable Velostat resistance. Therefore, parasitic current paths are unavoidable in this design since all nodes are always connected through the Velostat resistance so there is a backwards path through any node.

This reverse current flow can be prevented with the use of diodes (passive type) or transistors (active type). Diodes only allow current to pass in one direction. Figure 11 shows how the layout in Figure 10 can be modified. The diode protecting switch **e** prevents current flowing through it, and switch **s** is no longer mistakenly detected.

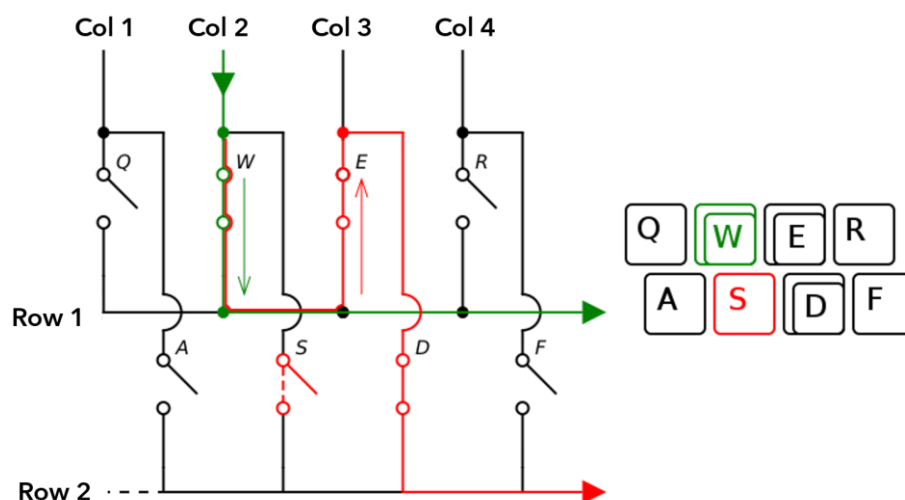


Figure 10 Basic schematic of a computer keyboard showing parasitic current paths can cause erroneous outputs[16]

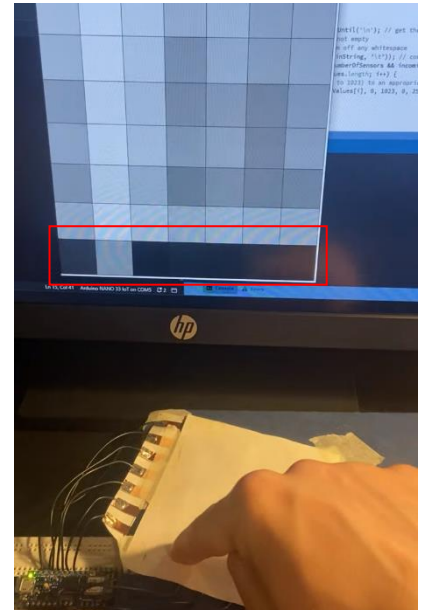


Figure 9 Demonstration of ghosting in the prototype pressure sensor array

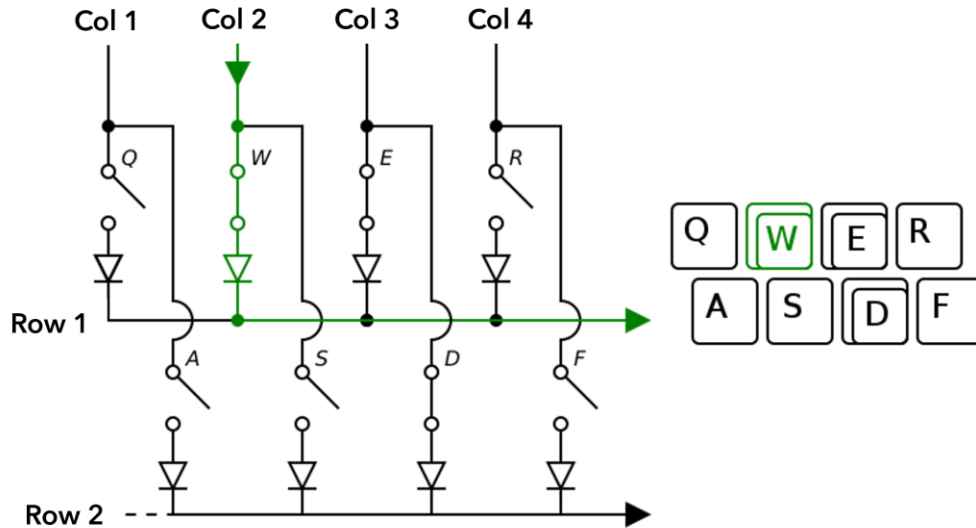


Figure 11 Modified schematic using diodes to prevent reverse parasitic current paths[16]

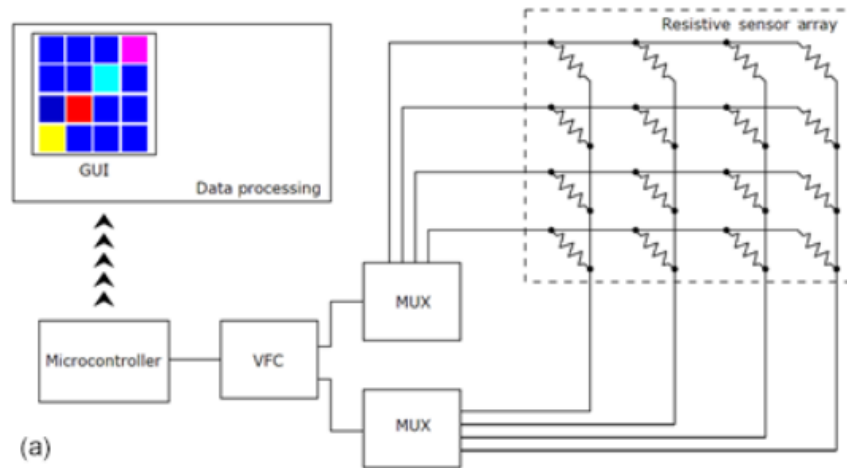


Figure 12 Modified system schematic includes Multiplexers (MUX) which can switch a single microcontroller input to multiple columns/rows[18]

2.2.2 Addressing the Matrix

As the size of the matrix scales even further it will become impractical to connect each row and column to an individual pin on a microcontroller (the Arduino Nano only has 8 Analog Input channels, so the prototype is already close to the limit). A solution to is to use multiplexers to digitally switch which row the current is applied to and which column is measured. A single cheap (~£2) multiplexer can switch a single analog input pin to 16 outputs[19] - allowing much larger sensor arrays to be created based on the same microcontroller.

Linked to the need to accommodate larger arrays without loss of function is the speed of addressing the matrix. Performance is acceptable in the 7x7 matrix presented here, but scaling this up to 50 or more rows and columns will lead to slower refresh rates [8]. The `Arduino AnalogRead()` command takes $\sim 110\mu\text{s}$ to run, which gives a theoretical refresh rate of 180Hz for a 49-element array. A 25x25 array on the other hand can only achieve a maximum of 14Hz, and a 150x100 array has a minimum of 0.6Hz.

In the context of the IPRD project, refresh rate is not the most crucial aspect of a pressure sensor but there are some simple steps that can be implemented to mitigate the loss of performance.

One method is to change the internal ADC (Analog-Digital Converter) clock speed. By default this is set to 16MHz divided by a prescale factor of 128, to give a default clock speed of 128kHz. One conversion using *analogRead()* takes 13 clock pulses to give a default sample rate of 9600Hz.[20]

Code in Appendix A changes the prescale factor from 128 to 16, increasing the sample rate by a factor of 8 to 75kHz (17 μ s per *analogRead()*), so a 150x100 array would have a refresh rate of 5Hz.

Another option is to use a dedicated external ADC IC. These are widely available and while they introduce some complexity and extra cost to the system their performance is far beyond the on-board Arduino ADC.

Figure 13 shows an example of an ADC chip costing <£5, with a maximum sample rate of 500kHz, and 16-bit resolution [21].

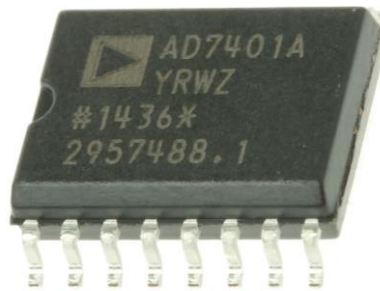


Figure 13 Discrete ADC IC [21]

3 Material Selection

Material choice for the initial prototype was not considered as it was intended to demonstrate proof-of-concept. However, the context of the IPRD project presents several key requirements for material selection that have to be considered for further development.

- **Unobtrusive design:** The pressure sensor cannot contain any rigid objects or components that would potentially introduce pressure points.
- **Flexible surface:** The IPRD is an air vessel that will inflate and deflate with the pressure sensor array between it and the patient. The sensor array should conform to the IPRD surface as much as possible.
- **Sterilisable:** Ultimately the pressure sensor will be used with ICU patients. It is completely unsustainable to make the sensor array single-use so there must be considerations allowing sterilisation/protection for reuse.

The paper and copper tape prototype largely fulfils the first two points but is unsuitable based on the third point and would not be strong enough to withstand repeated use.

As discussed in Mini-study 1, the most unobtrusive sensor architectures are piezoresistive and capacitive sensors. The paper prototype above has a thickness of just 0.8mm, and resistive and capacitive arrays can be manufactured flat, with no rigid objects to <1mm thickness[22][23].

The next major consideration is integrating the pressure sensor array with the IPRD air vessel. Figure 14 demonstrates for a row of parallel connected inflatable channels how the shape changes as they are pressurised. In the simplest case, two sheets can be fused together along two straight parallel lines of width d , leaving a channel of unfused material of width w . It is assumed that the sheet material is approximately inextensible, the channel is significantly longer than it is wide, and that its ends have been sealed.[24]

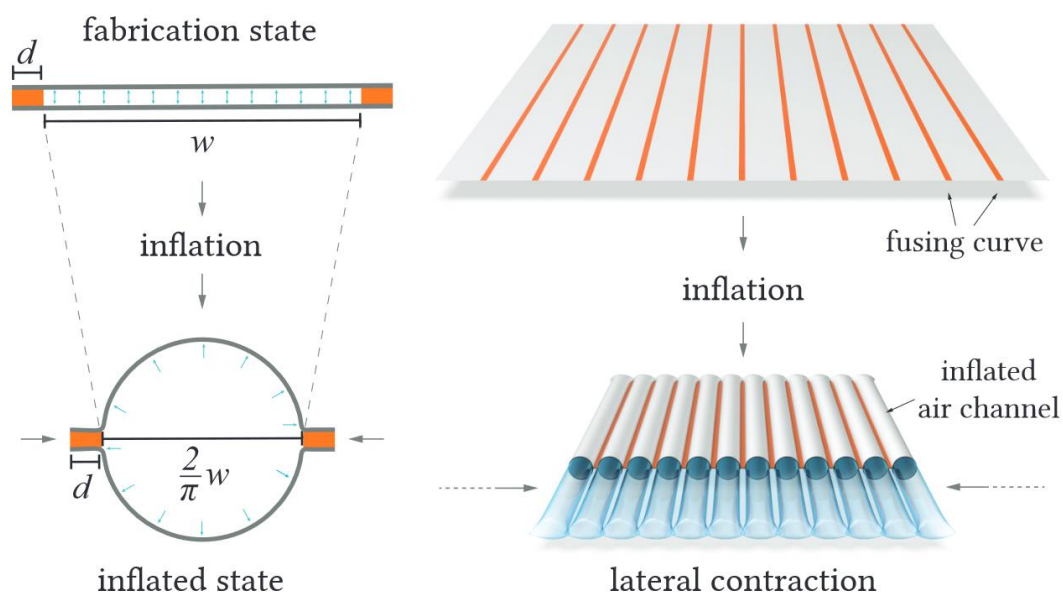


Figure 14 Air channels maximize volume when inflated by assuming a circular cross section. Consequently, the width of the vessel contracts [24]

Inflation pushes the sheets apart to maximise internal volume, so the inflated air channel approximates a cylinder when fully inflated. Channel width therefore shrinks from w to $\frac{2}{\pi}w$. Including the fusing seam (which does not contract), the contraction across the width, Δ , is:

$$\Delta = (w + d) - \frac{\frac{2}{\pi}w + d}{w + d}$$

Siéfert et al. 2020 [24][25] covers in detail the mechanical properties of inflatable tubular structures.

Figure 15 shows an approximation of the IPRD inflatable (provisional design as of December 2023), constructed from 3 separate air pockets. Using a squared central section prevents some of the lateral contraction shown in Figure 14, but the semicylindrical sections at either end do flatten into longer pieces (by a factor of $\frac{\pi}{2}$).

Since the cross-section of the IPRD vessel is constant along its width (all seams are straight and parallel) there is no out-of-plane deformation, or wrinkling expected where the pressure sensor is placed. This is beneficial since wrinkling or stretching the pressure sensor may cause erroneous readings. It also means the pressure sensor only has to be flexible and not stretchy, making material selection of the sensor less restricted.

Figure 16 shows an inflatable with concentric air pockets to demonstrate out-of-plane deformation that can occur and would make attaching a pressure sensing array difficult.

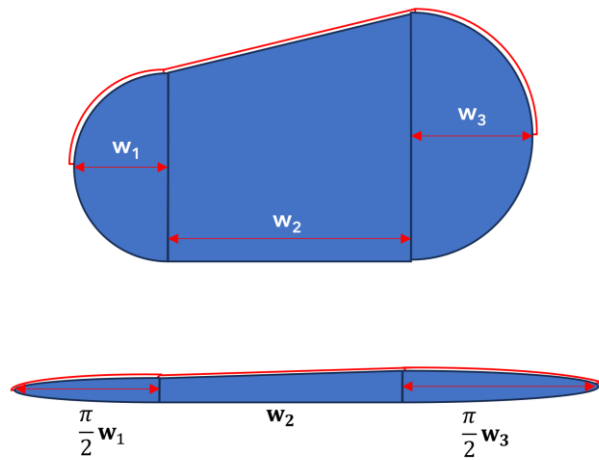


Figure 15 Approximate IPRD air vessel prototype design; Top:inflated cross section, Bottom:deflated. Red line indicates pressure sensing surface



Figure 16 Compound deformation caused by non-parallel seams can create surprising inflated shapes[24]

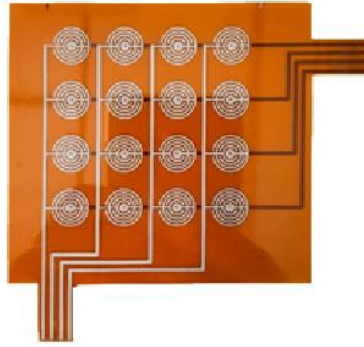


Figure 17 Silver ink screen-printed on Kapton sheet[18]

A thin and flexible sensor array is required that can be integrated with the IPRD as described previously. To replace the paper and copper tape used in the initial prototype, one option is to use a screen-printed flexible circuit. Figure 17 shows how this could be constructed using a Kapton (polyimide) substrate with conductive silver ink, to give a thickness of 125 μ m[18]

To replace the Velostat which can be unreliable and has widely varying material characteristics between samples, polydimethylsiloxane (PDMS) can be synthesised with varying concentrations of CNTs to achieve the same piezoresistive properties seen in Velostat, except with a greater level of control and reliability.[27][28]. Costa and Choi, 2017 [27], develop this further by using a screen-printing transfer process to deposit conductive ink onto a PDMS substrate. This is beneficial over the plastic film required for basic screen printing (Figure 17) as PDMS is not as prone to creasing or wrinkling and is softer and more flexible than plastic films.

4 Conclusions

A prototype of a thin, flexible piezoresistive-based pressure sensor array has been designed, constructed and tested. This was constructed from basic components (paper, copper tape, Velostat), to demonstrate proof-of-concept, and highlight key areas for further development, in line with the Design Thinking methodology which encourages using insight generated from prototyping and testing to further define the problem.

The insight from prototyping and testing highlighted software optimisation, sensor electronics and material selection as areas that could be improved/researched more for future prototype iterations.

The basic prototype functioned successfully using just an Arduino microcontroller, and some basic electronic components. Improvements to the hardware such as multiplexing to increase array size, and external ADCs to increase accuracy and speed will be explored in future prototyping activities to support eventual scaling-up of the design to a full-size array.

Improvements to the sensor architecture and materials were also highlighted as key areas requiring improvement to increase the reliability and durability of the sensor array. For example, screen-printing conductive ink to form the electrodes, and exploring implementing biocompatible elastomers (PDMS) as substrates could be key areas of future development with the ultimate goal of integrating a pressure sensing array into the surface of the IPRD air vessel.

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Appendix A: Arduino speed GPIO reading

```
#define cbi(sfr, bit) (_SFR_BYTE(sfr) &= ~_BV(bit))
#define sbi(sfr, bit) (_SFR_BYTE(sfr) |= _BV(bit))

void setup()
{
    sbi(ADCSRA, ADPS2);
    cbi(ADCSRA, ADPS1);
    cbi(ADCSRA, ADPS0);
    ...
}
```

Figure 18 Addition to Arduino script to change default sample rate scaling from 128 to 16

The above code [20] modifies the pre-scale factor from 128 to 16 to give an 8x increase in sample speed and therefore matrix refresh rate. There was not time to test this with the existing prototype, but future prototypes may include this (along with external ADCs or multiplexers to increase performance).